Comparison of Final Analysis Data Sets

Purpose

This report compares the original `final_data.csv` with `final_data_codex.csv`. Both files are read from `data/processed/` and are compared on their structure, country-year coverage, and shared variables.

```
library(dplyr)
library(readr)
library(tidyr)

original_data <- read_csv("../data/processed/final_data.csv", show_col_types = FALSE)
codex_data <- read_csv("../data/processed/final_data_codex.csv", show_col_types = FALSE)

key_columns <- c("iso", "year")
```

Data-set overview

The table below reports the number of rows, columns, and distinct country-year observations in each file. A duplicate count of zero means that every row has a unique `iso-year` combination.

```
overview <- bind_rows(
  original_data |> summarise(
    dataset = "final_data.csv",
    rows = n(),
    columns = ncol(original_data),
    unique_country_years = n_distinct(pick(all_of(key_columns))),
    duplicate_country_years = n() - n_distinct(pick(all_of(key_columns)))
  ),
  codex_data |> summarise(
    dataset = "final_data_codex.csv",
    rows = n(),
```

```
    columns = ncol(codex_data),
    unique_country_years = n_distinct(pick(all_of(key_columns))),
    duplicate_country_years = n() - n_distinct(pick(all_of(key_columns)))
  )
)

overview
```

```
# A tibble: 2 x 5
  dataset              rows columns unique_country_years duplicate_country_ye~1
  <chr>               <int>   <int>                <int>                   <int>
1 final_data.csv       3732      18                 3732                       0
2 final_data_codex.csv 3732      17                 3732                       0
# i abbreviated name: 1: duplicate_country_years
```

Variables present in each file

```
variable_comparison <- tibble(
  variable = union(names(original_data), names(codex_data)),
  in_final_data = variable %in% names(original_data),
  in_final_data_codex = variable %in% names(codex_data)
) |> arrange(variable)

variable_comparison
```

```
# A tibble: 18 x 3
   variable           in_final_data in_final_data_codex
   <chr>              <lgl>         <lgl>
 1 age_dep            TRUE          TRUE
 2 armed_conflict     TRUE          TRUE
 3 best               TRUE          FALSE
 4 drought            TRUE          TRUE
 5 earthquake         TRUE          TRUE
 6 gdp_1000           TRUE          TRUE
 7 infant_mortality   TRUE          TRUE
 8 iso                TRUE          TRUE
 9 male_edu           TRUE          TRUE
10 maternal_mortality TRUE          TRUE
11 neonatal_mortality TRUE          TRUE
```

```
12 oecd                TRUE          TRUE
13 pop_dens            TRUE          TRUE
14 rainfall_1000       TRUE          TRUE
15 temp                TRUE          TRUE
16 under5_mortality    TRUE          TRUE
17 urban               TRUE          TRUE
18 year                TRUE          TRUE
```

Both files now contain the binary `armed_conflict` variable. The only additional field in `final_data.csv` is `best`, the total battle-related deaths used to construct that binary exposure.

Country-year coverage

```
original_keys <- original_data |> distinct(across(all_of(key_columns)))
codex_keys <- codex_data |> distinct(across(all_of(key_columns)))

coverage_comparison <- tibble(
  comparison = c(
    "Country-years only in final_data.csv",
    "Country-years only in final_data_codex.csv",
    "Country-years in both files"
  ),
  n_country_years = c(
    nrow(anti_join(original_keys, codex_keys, by = key_columns)),
    nrow(anti_join(codex_keys, original_keys, by = key_columns)),
    nrow(inner_join(original_keys, codex_keys, by = key_columns))
  )
)

coverage_comparison
```

```
# A tibble: 3 x 2
  comparison                                    n_country_years
  <chr>                                                   <int>
1 Country-years only in final_data.csv                        0
2 Country-years only in final_data_codex.csv                  0
3 Country-years in both files                              3732
```

Shared-variable values

For every variable found in both files, the next table counts matching and non-matching values among country-years present in both files. Two missing values are treated as a match.

```
shared_value_columns <- setdiff(
  intersect(names(original_data), names(codex_data)),
  key_columns
)

joined_data <- inner_join(
  original_data |> select(all_of(c(key_columns, shared_value_columns))),
  codex_data |> select(all_of(c(key_columns, shared_value_columns))),
  by = key_columns,
  suffix = c("_original", "_codex")
)

value_comparison <- lapply(shared_value_columns, function(variable) {
  original_value <- joined_data[[paste0(variable, "_original")]]
  codex_value <- joined_data[[paste0(variable, "_codex")]]
  matches <- (is.na(original_value) & is.na(codex_value)) |
    (!is.na(original_value) & !is.na(codex_value) & original_value == codex_value)

  tibble(
    variable = variable,
    matching_values = sum(matches),
    differing_values = sum(!matches)
  )
}) |> bind_rows() |> arrange(desc(differing_values), variable)

value_comparison
```

```
# A tibble: 15 x 3
   variable           matching_values differing_values
   <chr>                        <int>            <int>
 1 armed_conflict                1036             2696
 2 age_dep                       3732                0
 3 drought                       3732                0
 4 earthquake                    3732                0
 5 gdp_1000                      3732                0
 6 infant_mortality              3732                0
 7 male_edu                      3732                0
```

```
 8 maternal_mortality           3732               0
 9 neonatal_mortality           3732               0
10 oecd                         3732               0
11 pop_dens                     3732               0
12 rainfall_1000                3732               0
13 temp                         3732               0
14 under5_mortality             3732               0
15 urban                        3732               0
```

Conclusion

The files have the same country-year coverage and share the binary `armed_conflict` exposure. Use `final_data.csv` when the underlying battle-death total (`best`) is also needed; otherwise, either file contains the same shared analysis variables. This report can be re-rendered whenever either CSV is regenerated, so its tables will update automatically.